1.35 Differentiation: Optimisation

Worked Solutions

1a)

$A = x^2$, $P = 4x$.

1b)

$A = 2x^2$, $P = 6x$.

1c)

$A = \pi x^2$, $P = 2\pi x$.

1d)

$A = x^2$,

$$P = x + \frac{\sqrt{17}}{2}x + \frac{\sqrt{17}}{2}x = (1 + \sqrt{17})x.$$

2a)

$V = x^3$, $A = 6x^2$.

2b)

$V = 3x^3$, $A = 4 \times 3x^2 + 2x^2 = 14x^2$.

2c)

$V = \pi x^2 h$, $A = 2\pi x^2 + 2\pi xh$.

2d)

$$V = \frac{1}{2}x^2 h, \quad A = 2 \times \frac{1}{2}x^2 + xh + 2\sqrt{\left(\frac{1}{2}x\right)^2 + x^2}$$
$$= x^2 + xh + \sqrt{5}x^2 = x^2 + (1 + \sqrt{5})xh.$$

3a)

$\dfrac{dV}{dr} = 20 - \dfrac{1}{2}r.$

3b)

$P = t^{-1} + \dfrac{1}{2}t$ so $\dfrac{dP}{dt} = -t^{-2} + \dfrac{1}{2}.$

3c)

$A = 2r - 10r^{-2}$ so $\dfrac{dA}{dr} = 2 + 20r^{-3}.$

3d)

$C = 10x^{-1} + \frac{1}{4}x^2$ so $\frac{dC}{dx} = -10x^{-2} + \frac{1}{2}x$.

4a)

$$25\pi = 4\pi r^2 \Rightarrow 25 = 4r^2 \Rightarrow r^2 = \frac{25}{4} \Rightarrow r = \frac{5}{2}.$$

4b)

$$\frac{1}{4} = \frac{144}{x^2} \Rightarrow x^2 = 576 \Rightarrow x = 24.$$

4c)

$$\frac{1}{2}x = \frac{10}{x^2} \Rightarrow x^3 = 20 \Rightarrow x = \sqrt[3]{20}.$$

4d)

$$\frac{1}{4}\pi r = \frac{27\pi^2}{r^2} \Rightarrow r^3 = 108\pi \Rightarrow r = \sqrt[3]{108\pi}.$$

4e)

$$\frac{1}{5}\pi r = \frac{3}{r^2} \Rightarrow r^3 = \frac{15}{\pi} \Rightarrow r = \sqrt[3]{\frac{15}{\pi}}.$$

5a)

Length of enclosure: $30 - 2x$.

Area: $A = x(30 - 2x)$.

5b)

$$\frac{dA}{dx} = 30 - 4x \quad \text{stat pt when } \frac{dA}{dx} = 0$$

$$30 - 4x = 0 \text{ so } x = \frac{30}{4}$$

$$A = 30\left(\frac{30}{4}\right) - 2\left(\frac{30}{4}\right)^2 = \frac{225}{2} = 112.5 \text{ m}^2.$$

6a)

$$C = 1156v^{-1} + \frac{1}{9}v + 50$$

$$\frac{dC}{dx} = -1156v^{-2} + \frac{1}{9} \quad \text{stat pt when } \frac{dC}{dv} = 0$$

$$-\frac{1156}{v^2} + \frac{1}{9} = 0 \Rightarrow \frac{1156}{v^2} = \frac{1}{9}$$

$$\Rightarrow \frac{v^2}{1156} = 9 \Rightarrow v = 102 \text{ kmh}^{-1} \quad \text{(must be positive).}$$

6b)

$$C = \frac{1156}{102} + \frac{102}{9} + 50 = \frac{218}{3} = £72.67.$$

7a)

Surface area: $100 = 2xy + 8xy + 8x^2$.

$$y = \frac{100 - 8x^2}{10x} = \frac{10}{x} - \frac{4x}{5}$$

Volume:

$$V = 4x^2y = 4x^2\left(\frac{10}{x} - \frac{4x}{5}\right) = 40x - \frac{16}{5}x^3 \text{ as req.}$$

7b)

$$\frac{dV}{dx} = 40 - \frac{48}{5}x^2 \quad \text{stat pt when } \frac{dV}{dx} = 0$$

$$40 - \frac{48}{5}x^2 = 0 \Rightarrow \frac{48}{5}x^2 = 40 \Rightarrow x^2 = \frac{25}{6}$$

So $x = \pm\frac{5}{\sqrt{6}}$ but x is a length.

$$V = 40\left(\frac{5}{\sqrt{6}}\right) - \frac{16}{5}\left(\frac{5}{\sqrt{6}}\right)^3 = 54.433105\ldots$$

So $V = 54.4$ cm^3 (1 DP).

7c)

$$x = \frac{5}{\sqrt{6}}, \quad 4x = \frac{20}{\sqrt{6}}, \quad y = 2\sqrt{6} - \frac{2}{3}\sqrt{6} = \frac{4}{3}\sqrt{6}$$

$\frac{5}{\sqrt{6}}$ by $\frac{20}{\sqrt{6}}$ by $\frac{4}{3}\sqrt{6}$.

8a)

Surface area: $48 = 2xy + 4xy + 2x^2$.

$$y = \frac{48 - 2x^2}{6x} = \frac{8}{x} - \frac{x}{3}$$

Volume:

$$V = 2x^2y = 2x^2\left(\frac{8}{x} - \frac{x}{3}\right) = 16x - \frac{2}{3}x^3 \text{ as req.}$$

8b)

$$\frac{dV}{dx} = 16 - 2x^2 \quad \text{stat pt when } \frac{dV}{dx} = 0$$
$$16 - 2x^2 = 0 \Rightarrow 2x^2 = 16 \Rightarrow x^2 = 8$$

So $x = \pm 2\sqrt{2}$ but x is a length.

$$V = 16(2\sqrt{2}) - \frac{2}{3}(2\sqrt{2})^3 = \frac{64}{3}\sqrt{2}$$

So $V = 30.2$ cm^3 (1 DP).

8c)

$$x = 2\sqrt{2}, \quad 2x = 4\sqrt{2}, \quad y = 2\sqrt{2} - \frac{2}{3}\sqrt{2} = \frac{4}{3}\sqrt{2}$$

$2\sqrt{2}$ by $4\sqrt{2}$ by $\frac{4}{3}\sqrt{2}$.

9a)

Surface area: $48\pi = \pi r^2 + 2\pi rh$.

$$h = \frac{48\pi - \pi r^2}{2\pi r} = \frac{30}{r} + \frac{r}{2}$$

Volume:

$$V = \pi r^2 h = \pi r^2 \left(\frac{48}{r} + \frac{r}{2}\right) = 24\pi r - \frac{\pi}{2} r^3 \text{ as req.}$$

9b)

$$\frac{dV}{dx} = 24\pi - \frac{3}{2}\pi r^2 \quad \text{stat pt when } \frac{dV}{dx} = 0$$
$$24\pi - \frac{3}{2}\pi r^2 = 0 \Rightarrow \frac{3}{2}\pi r^2 = 24\pi \Rightarrow r^2 = 16$$

So $x = \pm 4$ but x is a length.

$$V = 24\pi(4) - \frac{\pi}{2}(4)^3 = 64\pi \text{ cm}^3.$$

10)

Capacity of tray:

$$C = x(40 - 2x)(40 - 2x) = 4x^3 - 160x^2 + 1600x$$
$$\frac{dC}{dx} = 12x^2 - 320x + 1600 \quad \text{stat pt when } \frac{dC}{dx} = 0$$
$$12x^2 - 320x + 1600 = 0 \Rightarrow 4(x - 20)(3x - 20) = 0$$

So $x = 20$ or $x = \frac{20}{3}$ but physically $2x \neq 40$.

$$V = 4\left(\frac{20}{3}\right)^3 - 160\left(\frac{20}{3}\right)^2 + 1600\left(\frac{20}{3}\right) = \frac{128000}{27}$$

So $V = 4740.7$ cm^3 (1 DP).

11a)

Volume of container: $432\pi = \pi r^2 h$.

Surface area of container: $A = 2\pi rh + 2\pi r^2$.

$$A = 2\pi r \times \frac{432\pi}{\pi r^2} + 2\pi r^2 = \frac{864\pi}{r} + 2\pi r^2$$

$$\frac{dA}{dr} = -864\pi r^{-2} + 4\pi r \quad \text{stat pt when } \frac{dA}{dr} = 0$$

$$-864\pi r^{-2} + 4\pi r = 0 \Rightarrow \frac{864}{r^2} = 4r$$

$$\Rightarrow r^3 = 216 \Rightarrow r = 6$$

When $r = 6$ cm, $h = \dfrac{432}{6^2} = 12$ cm.

11b)

Cost for each container: $12 \times 0.04 = £0.48$.

12)

Let the length AB be $2x$ so length AD is $f(x)$.

Area is $A = 2x(10 - x^2) = 20x - 2x^3$.

$$\frac{dA}{dx} = 20 - 6x^2 \quad \text{max when } \frac{dA}{dx} = 0$$

$$20 - 6x^2 = 0 \Rightarrow 6x^2 = 20 \Rightarrow x^2 = \frac{10}{3}$$

$$\Rightarrow x = \sqrt{\frac{10}{3}}$$

$$\therefore \text{ max area: } 20\left(\sqrt{\frac{10}{3}}\right) - 2\left(\sqrt{\frac{10}{3}}\right)^3 = \frac{40}{3}\sqrt{\frac{10}{3}} = 24.3 \text{ units}^2.$$

13a)

Volume: $144\pi = \pi r^2 h + \dfrac{2}{3}\pi r^3$.

$$\Rightarrow 144 - \frac{2}{3}r^3 = r^2 h \Rightarrow \frac{144}{r^2} - \frac{2}{3}r = h$$

Semi-sphere plus side of cylinder area:

$$A = 2\pi r^2 + 2\pi rh = 2\pi r^2 + 2\pi r\left(\frac{144}{r^2} - \frac{2}{3}r\right)$$

$$= 2\pi r^2 + \frac{288\pi}{r} - \frac{4}{3}\pi r^2 = \frac{2}{3}\pi r^2 + \frac{288\pi}{r} \text{ as req.}$$

13b)

$$\frac{dA}{dr} = \frac{4}{3}\pi r - \frac{288\pi}{r^2} \quad \text{stat pt when } \frac{dA}{dr} = 0$$

$$\frac{4}{3}\pi r - \frac{288\pi}{r^2} = 0 \Rightarrow \frac{4}{3}\pi r = \frac{288\pi}{r^2}$$

$$\Rightarrow \frac{4}{3}r^3 = 288 \Rightarrow r^3 = 216 \Rightarrow r = 6 \text{ cm.}$$

13c)

$h = \dfrac{144}{36} - \dfrac{12}{3} = 0$ so that means that the least amount of material needed would only be a semi-sphere.

14a)

Volume: $500 = \pi r^2 h$.

Costs:

$$\begin{aligned} C &= 0.07 \times 2\pi r^2 + 0.03 \times 2\pi r h \\ &= \frac{7}{50}\pi r^2 + 0.06\pi r \left(\frac{500}{\pi r^2}\right) = \frac{7}{50}\pi r^2 + \frac{30}{r}. \end{aligned}$$

14b)

$$\frac{dC}{dr} = \frac{7}{25}\pi r - \frac{30}{r^2} \quad \text{stat pt when } \frac{dC}{dr} = 0$$

$$\frac{7}{25}\pi r - \frac{30}{r^2} = 0 \Rightarrow \frac{7}{25}\pi r = \frac{30}{r^2}$$

$$\Rightarrow r^3 = \frac{750}{7\pi} \Rightarrow r = 3.2429315622\ldots$$

so $r = 3.24$ cm (3 SF).

$$h = \frac{500}{\pi r^2} = 15.33\ldots \text{ so } h = 15.1 \text{ cm (3 SF).}$$

14c)

$$C = \frac{7}{50}\pi(3.2429315622\ldots)^2 + \frac{30}{3.2429315622\ldots} = 13.8763\ldots$$

so minimum cost is 14 pence (to the nearest pence).

15a)

Length: $\sqrt{x^2 + x^2} = \sqrt{2x^2} = \sqrt{2}x$.

Area: $\sqrt{2}xy$.

Area of two slanted sides: $2\sqrt{2}xy$.

15b)

surface area ends: $2x^2$.

surface area sides:

$$2\sqrt{2}xy + 2xy = 2xy(\sqrt{2} + 1)$$

Volume of prism: $500(\sqrt{2} - 1) = x^2 y$.

so $y = \dfrac{500(\sqrt{2} - 1)}{x^2}$.

Total costs:

$$C = 0.02 \times 2x\frac{500(\sqrt{2} - 1)}{x^2}(\sqrt{2} + 1) + 0.04 \times 2x^2$$

$$C = 20x^{-1} + \frac{2}{25}x^2.$$

15c)

$$\frac{dC}{dx} = -20x^{-2} + \frac{4}{25}x \quad \text{stat pt when } \frac{dC}{dx} = 0$$

$$-20x^{-2} + \frac{4}{25}x = 0 \Rightarrow \frac{4}{25}x = \frac{20}{x^2}$$

$$\Rightarrow x^3 = 125 \Rightarrow x = 5$$

So $C = \frac{20}{5} + \frac{2}{25} \times 25 = £4 + £2 = £6$.

16a)

Perimeter: $16 = 2x + 4y + \frac{1}{4} \times 2\pi x$.

Area of rectangles:

$$2xy = 2x\left(\frac{16 - 2x - \frac{1}{2}\pi x}{4}\right) = 8x - \left(1 + \frac{\pi}{4}\right)x^2$$

Area of quarter-circle: $\frac{1}{4}\pi x^2$.

Cost:

$$C = p\left[8x - \left(1 + \frac{\pi}{4}\right)x^2\right] + 0.8p \times \frac{1}{4}\pi x^2$$

$$C = 8px - \left(1 + \frac{\pi}{4}\right)px^2 + \frac{1}{5}\pi px^2$$

$$C = 8px - \left(1 + \frac{\pi}{20}\right)px^2.$$

16b)

$$\frac{dC}{dx} = 8p - 2\left(1 + \frac{\pi}{20}\right)px \quad \text{stat pt when } \frac{dC}{dx} = 0$$

$$8p - 2\left(1 + \frac{\pi}{20}\right)px = 0 \Rightarrow 2\left(1 + \frac{\pi}{20}\right)px = 8p$$

$$\Rightarrow \left(1 + \frac{\pi}{20}\right)x = 4 \Rightarrow x = \frac{4}{(1 + \frac{\pi}{20})}$$

$$\Rightarrow x = \frac{80}{20 + \pi} \text{ m as required.}$$

16c)

$$8\left(\frac{80}{20 + \pi}\right) - \left(1 + \frac{\pi}{4}\right)\left(\frac{80}{20 + \pi}\right)^2 = 6.31906734\ldots$$

Rectangle total area is 6.32 m^2.

$$\frac{1}{4}\pi\left(\frac{80}{20 + \pi}\right)^2 = 9.38606086\ldots$$

Quarter-circle area is 9.39 m^2.

17a)

Volume: $350 = \frac{1}{2}r^2 \times 0.7h$.

So $h = \frac{1000}{r^2}$.

Surface area:

$$A = 2hr + 0.7rh + 2 \times \frac{1}{2}r^2 \times 0.7$$

$$A = 2.7rh + 0.7r^2$$

$$A = 2.7r\left(\frac{1000}{r^2}\right) + 0.7r^2 = \frac{2700}{r} + 0.7r^2.$$

17b)

$$\frac{dA}{dr} = -2700r^{-2} + 1.4r \quad \text{stat pt when } \frac{dA}{dr} = 0$$

$$-\frac{2700}{r^2} + 1.4r = 0 \Rightarrow 1.4r = \frac{2700}{r^2}$$

$$\Rightarrow r^3 = \frac{13500}{7} \Rightarrow r = \sqrt[3]{\frac{13500}{7}} = 12.4473980\ldots$$

So $r = 12.4$ cm (3 SF).

$$A = \frac{2700}{\left(\sqrt[3]{\frac{13500}{7}}\right)} + 0.7\left(\sqrt[3]{\frac{13500}{7}}\right)^2 = 325.369$$

$A = 325$ cm^2 (3 SF).

$$h = \frac{1000}{r^2} = 6.45420637\ldots \text{ so } h = 6.45 \text{ cm (3 SF)}.$$